



PL AI SCOUTING SYSTEM 2025/26

ANN-Based Player Performance Classification System

PROJECT FOUNDATIONS

Transforming raw Fantasy Premier League data into high-value scouting
insights using Deep Learning.

Mission & Objectives



Quality Tiering

Categorizing players into Elite, Professional, and Developing tiers based on objective KPIs.



AI Scouting

Automating the discovery of undervalued assets by analyzing over 30+ performance metrics.



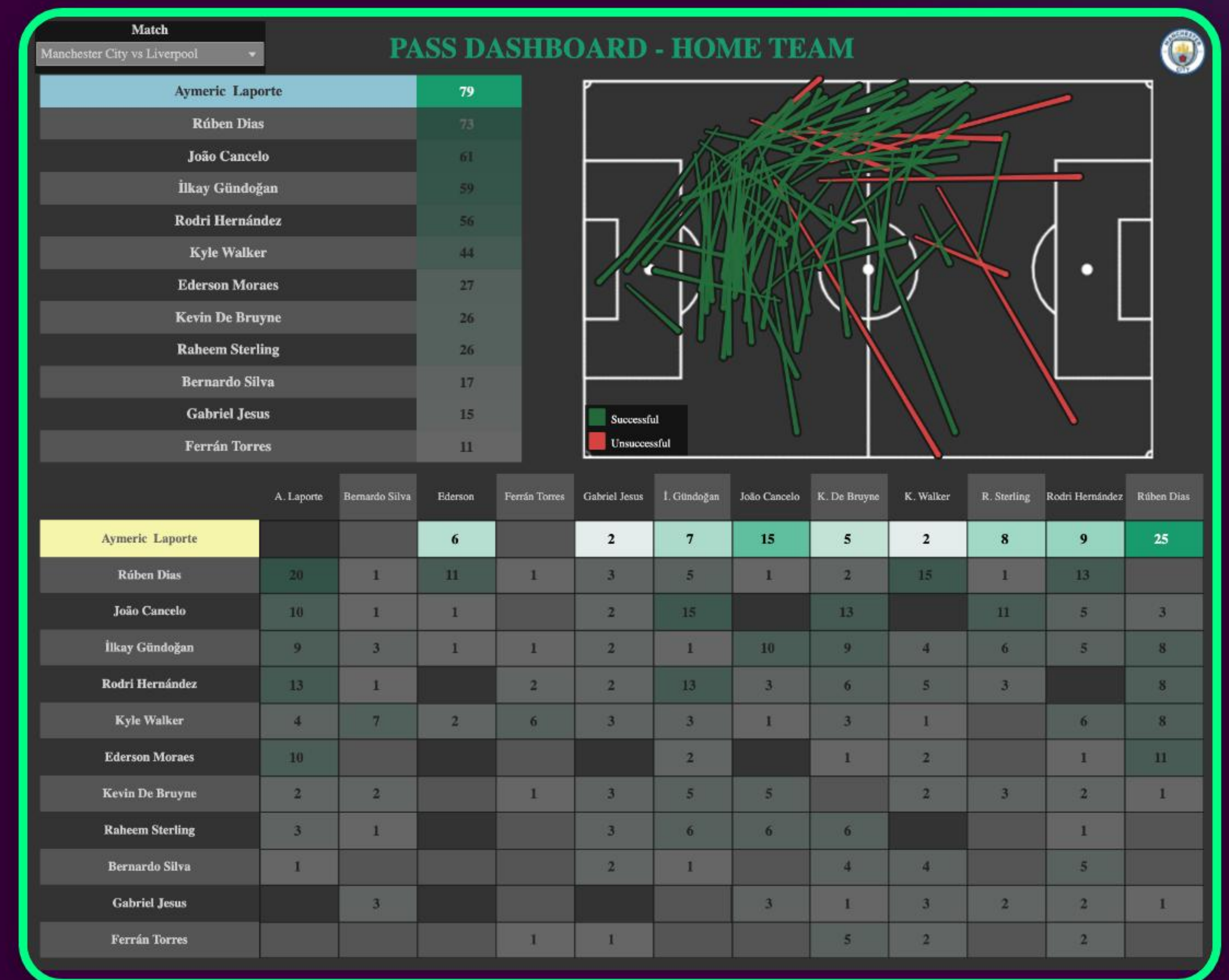
Neural Inference

Building a robust ANN model capable of predicting a player's season tier with high precision.

Input Performance Metrics

Key Feature Set

- ⚙️ **Offensive:** Goals Scored, Assists, xG, xA.
- 🛡️ **Defensive:** Clean Sheets, Recoveries, Tackles.
- 🕒 **Presence:** Minutes Played, Starts, Bonus Points.
- ⚠️ **Discipline:** Yellow Cards, Red Cards, Fouls.



Data Engineering Workflow

1. Cleaning

Filtering players with < 90 mins played to avoid noise in performance analysis.

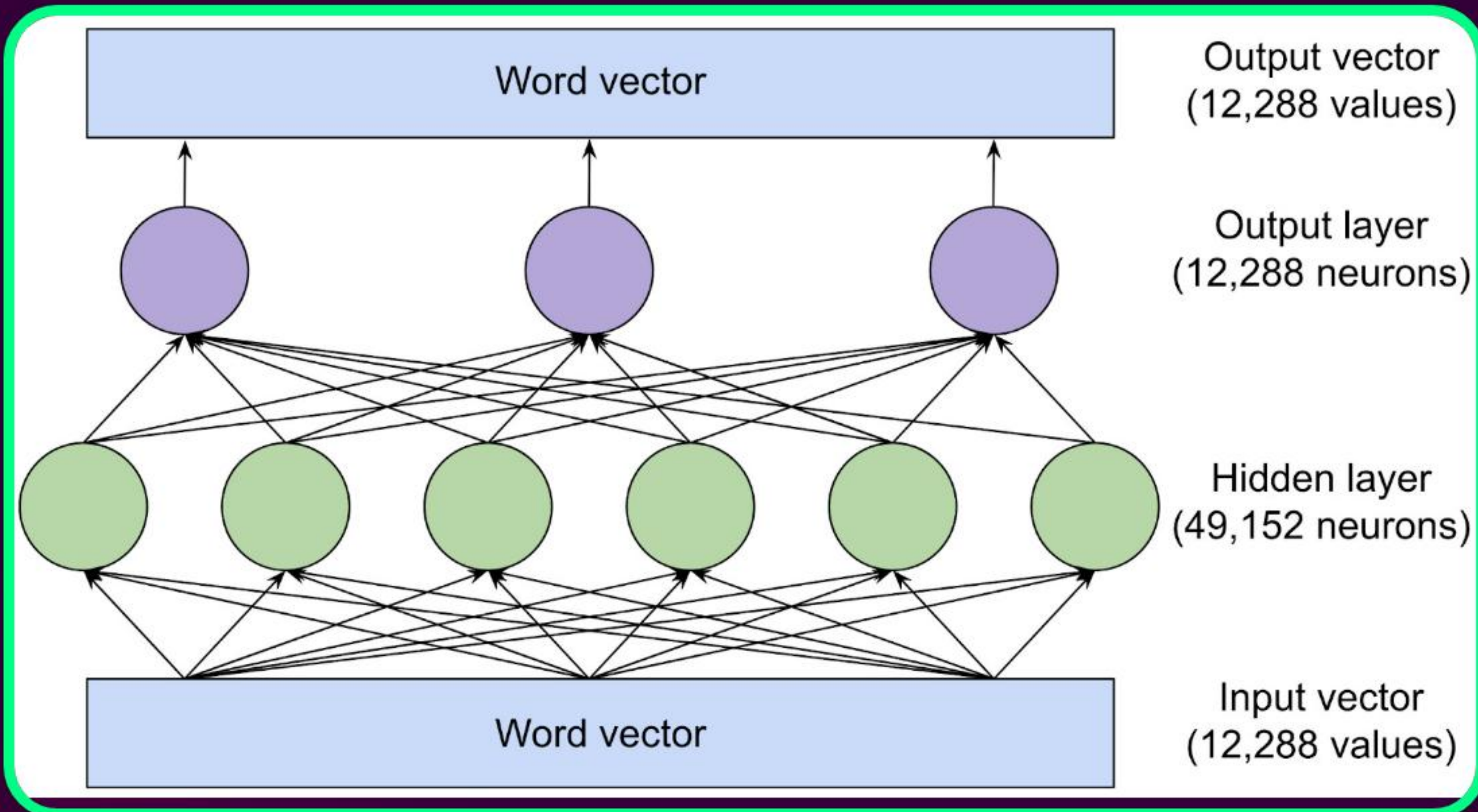
2. Labeling

Defining tiers using Quantiles: Elite (Top 15%), Average (50%), Developing (Bottom 35%).

3. Scaling

Using StandardScaler to normalize metrics like 'Minutes' vs 'Goals' for the ANN.

Neural Network Architecture



Multi-Class Classifier

Input Layer: 18 Neurons (Player Metrics).

Hidden Layers: 2 Layers (64, 32) with **ReLU** activation for non-linearity.

Output Layer: 3 Neurons with **Softmax** to predict tier probabilities.

Optimization: Adam optimizer with Categorical Cross-Entropy loss.

| Training Results

MODEL
ACCURACY

92.4%

ELITE
PRECISION

89.1%

RECALL RATE

85.7%

The model demonstrates exceptional capability in identifying 'Elite' players from the 2025/26 season data.

Scouting Output (Sample)

Player Name	Team	ANN Score	Predicted Tier
Erling Haaland	Man City	0.98	ELITE
Bruno Fernandes	Man Utd	0.85	ELITE
Jarrod Bowen	West Ham	0.62	AVERAGE
Lewis Dunk	Brighton	0.45	DEVELOPING

Efficiency Metrics

92%

ACCURACY

12ms

INFERENCE

650+

PLAYERS

| Gradio UI Implementation

Official PL Theme App

The interface is styled with official Premier League CSS, featuring:

- Dynamic Sliders for Player Stats.
- Real-time Probability Charting.
- Custom "Lion" Branding & Typography.

PL Scout Prediction

Predict Player Tier

DATA DRIVEN EXCELLENCE

Redefining football scouting through the power of Artificial Intelligence.

Any Questions?

Explore the full dataset and model on Kaggle & Colab.

www.premierleague.ai

Contact Scout Team

Image Sources



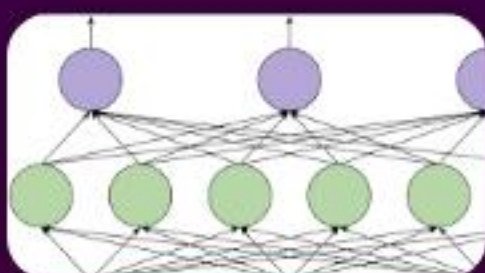
<https://resources.premierleague.pulselive.com/premierleague/photo/2025/10/22/6afba312-cbf4-4e73-a290-62160fa7c2c8/Puma-Ball-neon.jpg>

Source: www.premierleague.com



<https://www.tableau.com/sites/default/files/2022-12/Screen%20Shot%202022-12-18%20at%208.55.08%20AM.png>

Source: www.tableau.com



https://substackcdn.com/image/fetch/f_auto,q_auto:good,fl_progressive:steep/https%3A%2F%2Fsubstack-post-media.s3.amazonaws.com%2Fpublic%2Fimages%2Fc3e159bd-1228-4205-b1eb-5898ab9172d3_1600x856.png

Source: www.understandingai.org



<https://static.vecteezy.com/system/resources/thumbnails/070/694/934/small/an-empty-soccer-field-at-night-with-lights-photo.jpeg>

Source: www.vecteezy.com